Investigating Performance Under Varying Selective Pressures in Artificial Organisms

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Abstract

Evolution is the process of an organism changing its biology over generations [2]. Genetic algorithms mimic this process by allowing a certain number of 'organisms' to find an optimal solution to a problem, assessing their performances, selecting the successful ones, adding variation, and repeating until an optimal solution is reached [1]. These genetic algorithms have a variety of applications, being especially useful in the development of artificial intelligence, such as those in self-driving cars [6]. We determined the correlation between competition and evolutionary efficiency by running simulations with unique organisms with muscles and nodes that act on an oscillating cycle, evaluating how far each one goes, selecting varying percentages of the best performers to reproduce, and repeating for 50 generations. After averaging the data for each generation of each trial, the data suggested that there existed a moderately strong correlation between the percentage of organisms that survived and the average distance walked. As the rate of survival decreased, the speed at which organisms learned to walk further distances increased, but not the maximum distance in a generation, or the final solution. The results of ANOVA testing concluded our results to be statistically significant. These results align with our current understanding of the effect of selective pressure on evolution, that the levels of selective pressure must justify the decrease in variance and population [5].

1 Introduction

Evolution is the process by which organisms' biology changes over generations [2]. These changes come from slight genetic variations, caused by mutations during reproduction [2]. The process of evolution is made possible by these mutations and the existence of competition [3]. Organisms that mutate with favorable traits will in turn survive longer and produce more offspring, passing on the traits to the next generation [3]. Over many generations, sometimes spanning millions of years, subtle mutations morph into species-wide changes, or sometimes the creation of a new species entirely [3]. This notion of evolution was discovered in part by Charles Darwin, which is why the theory has been coined "Darwinism" [3].

Evolutionary design by computers mirrors the fundamentals of natural evolution, or Darwinism, to simulate natural evolutionary processes in biology. At its core, it utilizes algorithms based on selection, mutation, and recombination to iteratively enhance designs based on certain criteria or predefined functions [1]. By having organisms randomly attempt to find the optimal solution, reproducing the successful ones while killing off the rest, and adding a slight variation to their behavior between generations, the population "evolves" toward an optimal solution (Figure 1).

These genetic algorithms are used in a variety of technical, scientific, and professional fields to optimize the efficiency of programs or technology [8]. An example of this is when engineering self-driving cars, such as by companies like Tesla. By using the techniques described above, the program can "evolve" to be more efficient at finding paths, more human-like in its driving, and safer or more aware of its surroundings [6].

The effect of competition on a species' ability to evolve has been the topic of many studies. The common conclusion is that the existence of competition is necessary and beneficial to a point, but the benefits must increase selective pressure enough to justify the decrease in abundance [5]. Too little or too much competition and the organism will fail to properly evolve or converge on a non-ideal solution. The latter can be seen when studying the adaptation of tadpoles to invasive crawfish. While typically tadpoles vary in the time it takes for them to develop, the addition of harsh selective pressure in the form of crawfish led the tadpoles to adapt to develop much quicker [4]. While tadpoles with this adaptation were more likely to survive the competitive environment, the quicker development led to defects such as smaller body size and poorer jumping ability [4].

The purpose of this experiment is better to understand the correlation between competition and evolutionary efficiency. By running these simulations with different percentages of organisms being killed after each generation, we can find the optimal level of competition, which can be extrapolated to understand how real-world competition influences the speed and proficiency of evolution. We hypothesize that the ideal percentage of organisms killed after each generation will be 50 percent. If too many organisms are killed after each generation, the species will converge sooner, reaching a less preferable end product. If not enough organisms are killed after each generation, the process of evolution will occur too slowly, and the species will not improve over the limited generations given.

2 Methods

2.1 Environmental Setup and Control Group

We began by constructing an algorithm that generated an organism with interconnected nodes and muscles (Figure 2). Each organism possesses a central node from which all other node positions are relative. Nodes are defined by friction, mass, and position relative to the central node, with a small chance of additional node generation or deletion during creation (Figure 3). Node friction ranged from 0 (minimum) to 5 (maximum), with a mutation rate of 0.1. Node mass ranged from 0.5 (minimum) to 4 (maximum), with a mutation rate of 0.001. The initial position of nodes was offset from the central node by X=4 and Y=2, with a mutation range of 0.01 in both the X and Y directions.

Muscles, connecting every node pair, are characterized by default length, contracted length, cycle length, cycle contracted percent, cycle contracted offset, and strength. Muscle strength was fixed at 1 with no mutation. Muscle full length ranged from 2 (minimum) to 6 (maximum), with a mutation rate of 0.001. Muscle contracted length was set to 2, with a mutation rate of 0.0005. Muscle cycle length ranged from 0.05 (minimum) to 4 (maximum), with a mutation rate of 0.0005. The muscle cycle contracted percentage was set to 1, with no mutation. Muscle cycle contracted offset was set to 0, with no mutation. Muscle cycle offset ranged from 0 (minimum) to 2 (maximum), with a mutation rate of 0.0001. Muscle behavior was simulated as simple harmonic oscillators.

800 organisms were then placed on a flat surface and given 15 seconds to move in a constant direction. The average position of all nodes was calculated to determine the distance traveled. The organisms were ranked based on their distance traveled and all organisms were replicated to create the next generation. This process was repeated through 50 generations.

2.2 Experimental Groups

The steps above were repeated, except that instead of all organisms being replicated to create the next generation, only the top 10 percent, 25 percent, 50 percent, and 75 percent of organisms in the ranking were replicated to the next generation.

3 Results

We successfully ran five simulations (one per experimental group) with 50 generations per group. Our control group accurately displayed no correlation between generations and average distance walked under no selective pressure (Figure 4). We observed a moderately strong association between the percentage of organisms that survived and the average distance walked by organisms of subsequent generations (Table 1). Specifically, as the survival rate decreased (indicating a smaller percentage of offspring that proceeded to the next generation), the rate at which organisms learned to walk farther distances increased, but the overall distance traveled did not necessarily increase, exhibiting high levels of variability (Figure 5). This suggests that under stronger selective pressures, organisms may tend to learn at a faster rate but may not achieve the highest performance when compared to organisms under lower selective pressures. This is supported by the 25 percent, 50 percent, and 75 percent survival cutoff groups where there is a lower initial rate of distance walked but continued growth for groups with a higher cutoff compared to a lower cutoff with plateaued or began decreasing (Figure 6, 7, 8).

To assess whether the data collected between groups exhibited significant levels of difference, we performed a one-way analysis of variance (ANOVA) test with the following null and alternative hypotheses:

Based on the results of the ANOVA test, we found our results to be statistically significant because our p-value of 1.11841E-39 is less than our alpha level of 0.05 and F ¿ Fcrit (Table 2). As a result, we successfully reject the null hypothesis because we cannot attribute the variance between the experimental groups to random chance.

4 Discussion

In this experiment, the results do not fully support our initial hypothesis that the ideal percentage of organisms killed after each generation will be 50 percent to yield the best performance. While we were correct in predicting that organisms under too intense and too mild selective pressures would walk the farthest net positive distance, we failed to foresee that organisms under a milder selective pressure (75 percent survival cutoff) continued to learn and perform better while the performance of organisms under slightly more selective pressures (25 percent and 50 percent survival cutoff) began to decline. This is supported by the logarithmic regressions as the 75 percent survival cutoff group yielded the highest slope value of 5.6373 while having the highest correlation coefficient compared to the 25 percent survival cutoff group with a slope value of 4.4280 and the 50 percent survival cutoff group with the slope value of 3.0666 (Figure 6, 7, 8).

These results are somewhat consistent with previous studies surrounding how different selective pressures affect evolution and performance. As stated in the introduction, extreme levels of selective pressures by invasive crayfish on frog tadpoles were found to cause rapid evolutionary changes but reduced the size and jumping ability of young frogs in the long term as later generations were affected by this rapid development [4]. In this experiment, when artificial organisms were faced with the greatest selective pressure (10 percent survival cutoff), their ability to learn in the first ten generations was apparent through the

high rate of change. However, high levels of variation were observed in later generations which aligns with the negative influence that high selective pressures have on long-term performance (Figure 5). This is also seen in a study that found that stronger selection reduces the rate of evolution if recombination is the primary driver behind genetic variation which is similar to the variation of the nodes and muscles between generations in this study [7]. Overall, this experiment has shown the relationship between varying selective pressures and the respective performance as well as the potential of an optimal selective pressure. Through this, we can learn how to apply evolutionary algorithms in different contexts such as machine learning to find the most optimal solution in different or new environments.

The biggest obstacle we faced was deciding the most optimal mutation rates for muscles and nodes that would allow organisms to "learn" and improve across generations. This was overcome by trial and error, and observing the effect that certain ranges of parameters had on successive generations. This led us to the parameter settings that were ultimately used when simulating the control and experimental groups to ensure that we could achieve a balance between generation stability and generation growth where the artificial organisms could learn to walk without excessive variability or stagnation. To better simulate real-life conditions, future investigations surrounding this topic could involve increasing the type of movements that organisms must perform, including learning to navigate by having the ability to change directions in a multi-dimensional model. This will allow us to continue studying how organisms adapt to new conditions while increasing the complexity of their environment.

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